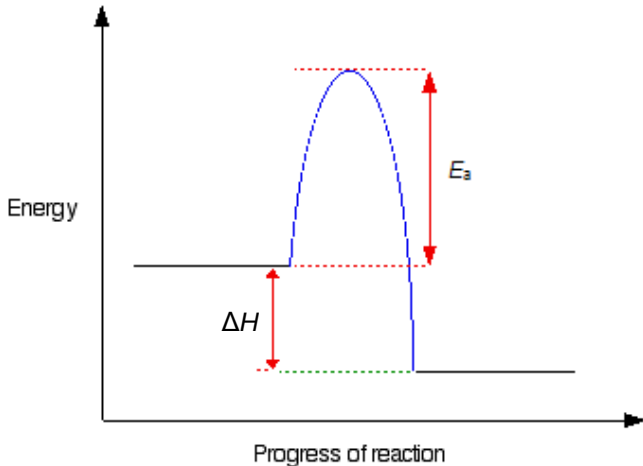


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)	(i)		(average) energy required to break / released by making (1) 1 mol of bonds (1)	2			2		
		(ii)		bonds broken $\text{C}\equiv\text{C} + 2(\text{H}-\text{H}) = 839 + 2(436) = 1711$ (1) bonds formed $(\text{C}-\text{C}) + 4(\text{C}-\text{H}) = 348 + 4(413) = 2000$ (1) $\Delta H = 1711 - 2000 = -289 \text{ kJmol}^{-1}$ (1) ecf possible		3		3	1	
		(iii)		 products at lower energy than reactants (1) ecf possible from part (ii) ΔH and E_a labelled (1)	2			2		